Blood Glucose Variability Factors

Factor	Category	Mechanism / what it changes	Primary glucose effect	Typical delay	Insulin resistance	Insulin sensitivity	Good data for confidence	Long-term learning	Automation Eligibility	Current pumps
Active energy burned	Activity	Load/intensity proxy	Usually lowers	Minutes to hours	—	Increases	2–3 weeks	Ongoing	No	No
Aerobic exercise	Activity	Improves sensitivity	Lowers	Immediate and 6–24h	—	Increases	2–4 weeks	Ongoing	No	Yes
Alcohol intake	Nutrition	Hepatic suppression	Raises then lowers	4–12h	—	—	1–2 months	Ongoing	No	No
Altitude	Environment	Stress/dehydration	Both	Hours to days	Increase	—	1–2 months	Episodic	No	No
Anaerobic / HIIT	Activity	Adrenaline response	Raises then lowers	Immediate + delayed	Increases (acute)	Increases (later)	2–4 weeks	Ongoing	No	No
Basal insulin delivered	Insulin delivery	Background glucose disposal and hepatic suppression	Both (too low raises, too high lowers)	Minutes to hours	—	—	2–4 weeks	Ongoing	Yes	Yes
Basal rate changes during day	Insulin delivery	Time-varying background insulin	Both	Minutes to hours	—	—	2–4 weeks	Ongoing	Yes	Yes
Body temperature	Physiology	Fever / cycle proxy	Raises when febrile	Hours to days	Increase	—	1–2 months	Ongoing	No	No
Bolus insulin (meal bolus)	Insulin delivery	Offsets meal glucose rise	Lowers if adequate	10–60 min onset, hours tail	—	—	2–3 weeks	Ongoing	Yes	Yes
Bolus stacking	Insulin (derived)	Overlapping doses	Lowers later	1–4 hours	—	—	2–4 weeks	Ongoing	Yes	Yes
Caffeine	Nutrition	Catecholamine release	Both	0–2 hours	Partial increase	—	1–2 months	Ongoing	No	No
Carbohydrates	Nutrition	Primary glucose input	Raises	10–90 min	—	—	2–3 weeks	Ongoing	No	Yes
CGM glucose readings	Glucose	Observed interstitial glucose	Measurement	5–15 min lag	—	—	7–14 days	Ongoing	Yes	Yes
CGM sensor lag	Glucose measurement	Interstitial vs blood delay	Apparent delay	Minutes	—	—	7–14 days	Ongoing	No	No
Circadian misalignment	Sleep	Hormonal rhythm shift	Both	Days	Increase	—	1–2 months	Ongoing	No	No
Cold exposure	Environment	Stress response	Can raise	Minutes to hours	Increase	—	1–2 months	Ongoing	No	No
Complex carbs	Nutrition	Slower absorption	Raises later	30–180 min	—	—	2–4 weeks	Ongoing	No	No
Compression lows	Glucose measurement	Mechanical pressure artifact	False low	Immediate	—	—	2–4 weeks	Ongoing	No	No
Correction bolus	Insulin delivery	Lowers hyperglycemia	Lowers	10–60 min onset	—	—	2–3 weeks	Ongoing	Yes	Yes
Cortisol (chronic stress)	Hormones / cycle	Hepatic glucose output	Raises	Minutes to hours	Increase	—	1–2 months	Ongoing	No	No
Extended / dual bolus	Insulin delivery	Distributes bolus over time	Lowers later	1–6 hours	—	—	2–4 weeks	Ongoing	Yes	Yes

Factor	Category	Mechanism / what it changes	Primary glucose effect	Typical delay	Insulin resistance	Insulin sensitivity	Good data for confidence	Long-term learning	Automation Eligibility	Current pumps
Fat (total)	Nutrition	Delayed gastric emptying	Raises later	2–6 hours	Increases	—	1–2 months	Ongoing	No	No
Fiber	Nutrition	Slows absorption	Delays spike	30–240 min	—	—	1–2 months	Ongoing	No	No
Fingerstick glucose	Glucose	Capillary glucose reference	Both	Immediate	—	—	7–14 days	Ongoing	No	Yes
Gaps in CGM data	Glucose measurement	Missing observations	Unknown	Immediate	—	—	Immediate	Ongoing	No	No
Glucose trend / rate of change	Glucose (derived)	Slope / acceleration	Both	Immediate	—	—	7–14 days	Ongoing	Yes	Yes
Growth hormone overnight	Hormones / cycle	Nighttime resistance	Raises	Night	Increase	—	1–2 months	Ongoing	No	No
Heart rate	Physiology	Stress/intensity proxy	Both	Immediate	Can increase	Can increase	1–2 weeks	Ongoing	No	No
Heart rate variability	Physiology	Recovery/stress proxy	Higher BG when low	Days	Increase when low	Increases when high	3–4 weeks	Ongoing	No	No
Heat exposure	Environment	Increased absorption	Can lower	Minutes to hours	—	Increases	1–2 months	Ongoing	No	No
Hormonal contraception	Medications	Hormonal modulation	Both	Weeks	Partial increase	Partial increase	3–6 months	Ongoing	No	No
Infection / illness	Illness	Inflammation-driven resistance	Raises	Hours to days	Increase	—	1–2 months	Episodic	No	No
Infusion site aging	Pump / set	Reduced absorption	Raises	Hours to days	—	—	1–2 months	Ongoing	No	No
Insulin on Board	Insulin (derived)	Active insulin remaining	Lowers / hypo risk	Minutes to hours	—	—	2–3 weeks	Ongoing	Yes	Yes
Insulin type (rapid vs ultra-rapid)	Insulin delivery	Action curve shape	Both	Minutes to hours	—	—	2–3 weeks	Ongoing	Yes	Yes
Late bolus timing mismatch	Insulin (derived)	Carbs before insulin	Raises then lowers	0–3 hours	—	—	2–4 weeks	Ongoing	Yes	Yes
Meal timing	Nutrition	Insulin–carb alignment	Both	Immediate	—	—	2–3 weeks	Ongoing	No	No
Menstrual follicular phase	Hormones / cycle	Increased sensitivity	Can lower	Days	—	Increases	3 months	Ongoing	No	No
Menstrual luteal phase	Hormones / cycle	Increased resistance	Raises	Days	Increase	—	3 months	Ongoing	No	No
Missed bolus	Behavior	Omitted insulin	Raises	0–3 hours	—	—	2–4 weeks	Ongoing	Yes	Yes
Occlusion / kink / leak	Pump / set	Failed delivery	Raises	Immediate to hours	—	—	Immediate	Episodic	No	No
Ovulation window	Hormones / cycle	Variability spike	Both	Days	—	Increases	3 months	Ongoing	No	No
Post-exercise sensitivity window	Activity	Enhanced insulin action	Lowers later	6–48 hours	—	Increases	1–2 months	Ongoing	No	No

Factor	Category	Mechanism / what it changes	Primary glucose effect	Typical delay	Insulin resistance	Insulin sensitivity	Good data for confidence	Long-term learning	Automation Eligibility	Current pumps
Protein	Nutrition	Gluconeogenesis	Raises later	2–6 hours	—	—	1–2 months	Ongoing	No	No
Puberty / adolescence	Hormones / lifecycle	Growth hormone surges	Raises baseline	Weeks to months	Increase	—	3–6 months	Ongoing	No	No
Recovery phase after illness	Illness	Sensitivity rebound	Lowers	Days	—	Increases	1–2 months	Episodic	No	No
Reservoir empty	Pump / set	No insulin delivered	Raises	Immediate	—	—	Immediate	Episodic	Yes	Yes
Respiratory rate	Physiology	Illness/recovery proxy	Both	Days	Increase (illness)	—	1–2 months	Ongoing	No	No
Resting heart rate	Physiology	Chronic stress marker	Raises if elevated	Days	Increase	—	3–4 weeks	Ongoing	No	No
Routine consistency	Behavior	Predictability reduces variance	Reduces variability	Days to weeks	—	Increases	1–2 months	Ongoing	No	No
Sensor drift / calibration error	Glucose measurement	Systematic bias	False high/low	Hours to days	—	—	2–4 weeks	Ongoing	No	No
Simple carbs	Nutrition	Fast absorption	Raises faster	5–45 min	—	—	2–4 weeks	Ongoing	No	No
Sleep duration	Sleep	Sleep debt increases resistance	Raises next day	Next day	Increase	—	3–4 weeks	Ongoing	No	Yes
Sleep fragmentation / quality	Sleep	Poor recovery	Raises next day	Next day	Increase	—	1–2 months	Ongoing	No	No
Steps / general movement	Activity	Mild uptake	Lowers mildly	Minutes to hours	—	Increases	2–3 weeks	Ongoing	No	No
Suspend / pause delivery	Insulin delivery	Stops insulin flow	Raises	Immediate to hours	—	—	2–3 weeks	Ongoing	No	Yes
Systemic steroids	Medications	Strong insulin resistance	Raises	Hours to days	Increase	—	3–4 weeks	Episodic	No	No
Target glucose setting	Pump / algorithm	Higher allowed BG	Raises baseline	Chronic	—	—	1–2 months	Ongoing	No	No
Temporary basal decrease	Insulin delivery	Short-term background reduction	Both	Minutes to hours	—	—	2–4 weeks	Ongoing	Yes	Yes
Temporary basal increase	Insulin delivery	Short-term background increase	Both	Minutes to hours	—	—	2–4 weeks	Ongoing	Yes	Yes
Thyroid medication/status	Medications	Metabolic rate changes	Both	Weeks	Increase or decrease	Increase or decrease	3–6 months	Ongoing	No	No
Water intake / hydration	Nutrition	Indirect physiology	Raises if dehydrated	Hours	Partial increase	—	1–2 months	Ongoing	No	No
Workout session (any)	Activity	Increases uptake; hormonal effects	Both	Immediate to 48h	—	Increases	2–3 weeks	Ongoing	No	Yes